

Project Design Phase

Solution Architecture

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Team ID: SWTID-2026-5226

Project Name: Study Buddy API

Maximum Marks 4 Marks

Solution Architecture:

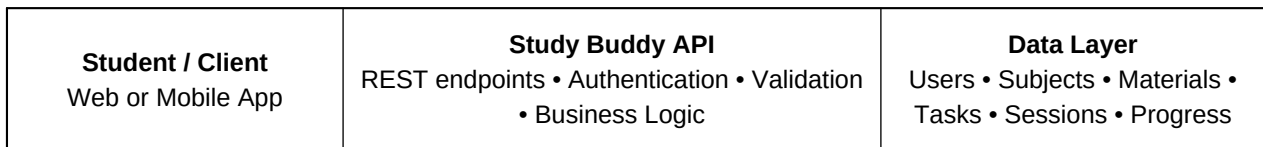
Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Study Buddy API – Proposed Solution Architecture

The Study Buddy API provides a backend service for student study activities. It exposes secure API endpoints for user accounts, study materials, study sessions, tasks, and progress tracking. A client application communicates with the API, while the API validates requests, applies business rules, stores data, and returns structured responses.

Example – Solution Architecture Diagram:



Data flow: Client → API request → Authentication & validation → Business logic → Database → API response → Client

Figure 1: Study Buddy API solution architecture and high-level data flow

Study Buddy API – Data Flow Diagrams

DFD Level 0 – Context Diagram

Student		Study Buddy API		Database
Study requests	→	Authenticate • Process • Manage study data	→	Store / retrieve data
Study data / credentials	←	API responses / progress	←	Requested records

Figure 2: Context-level data flow for the Study Buddy API

DFD Level 1 – Core API Processes

1. User Authentication Register / Login / Token validation	2. Study Management Subjects / Materials / Tasks
3. Study Session Management Start / update / complete sessions	4. Progress Tracking Calculate and return study progress

Typical flow: Student → Authentication → Study Management / Sessions → Progress Tracking → Database → Response

Figure 3: Level 1 decomposition of Study Buddy API processes

Study Buddy API – User Stories

ID	User Story
US-01	As a student, I want to register and log in so that I can securely access my study data.
US-02	As a student, I want to create and manage subjects so that I can organize my studies.
US-03	As a student, I want to add and view study materials so that I can keep learning resources in one place.
US-04	As a student, I want to create, update, and complete study tasks so that I can manage my study plan.
US-05	As a student, I want to start and complete study sessions so that my study activity can be recorded.
US-06	As a student, I want to view my progress so that I can understand how much of my study plan I have completed.
US-07	As a student, I want the API to protect my account and study data so that only authorized users can access them.

Figure 4: User stories for the Study Buddy API

Key API Components

- **API Layer:** REST endpoints for authentication, subjects, materials, tasks, sessions, and progress.
- **Authentication:** Credential validation and token-based authorization for protected endpoints.
- **Business Logic:** Validation of study rules and processing of user actions.
- **Data Storage:** Persistent storage for user profiles, study content, tasks, sessions, and progress.
- **Response Layer:** Structured success and error responses returned to the client.

Reference: Template supplied by the project team; architecture and Study Buddy API-specific content prepared from the requested project details.